Olympiad Circle Geometry, Power of a Point & Radical Axes Practice Exercises

1 In the circle shown, chords AB and CD intersect internally at point P. Given $PA = 4$, $PB = 9$, and $PC = 3$, find the exact segment length PD.

a. 10

b. 14

c. 12

d. 9

2 In the circle shown, chords AB and CD intersect internally at point P. Given $PA = 6$, $PB = 8$, and $PC = 4$, find the exact segment length PD.

a. 10

b. 12

c. 14

d. 9

3 In the circle shown, chords AB and CD intersect internally at point P. Given $PA = 5$, $PB = 12$, and $PC = 6$, find the exact segment length PD.

a. 10

b. 11

c. 12

d. 7

4 In the circle shown, chords AB and CD intersect internally at point P. Given $PA = 6$, $PB = 10$, and $PC = 5$, find the exact segment length PD.

a. 11

b. 14

c. 9

d. 12

5 Cyclic quadrilateral $ABCD$ (shown schematically) has consecutive side lengths $AB = 3,\ BC = 4,\ CD = 3,\ DA = 4$, and diagonal $AC = 5$. By Ptolemy's Theorem, find the exact length of the other diagonal BD.

a. $\frac{7}{5}$

b. $\frac{24}{5}$

c. 5

d. 7

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6 Cyclic quadrilateral $ABCD$ (shown schematically) has consecutive side lengths $AB = 5,\ BC = 12,\ CD = 5,\ DA = 12$, and diagonal $AC = 13$. By Ptolemy's Theorem, find the exact length of the other diagonal BD.

a. $\frac{119}{13}$

b. 13

c. $\frac{120}{13}$

d. 15

7 Cyclic quadrilateral $ABCD$ (shown schematically) has consecutive side lengths $AB = 6,\ BC = 8,\ CD = 6,\ DA = 8$, and diagonal $AC = 10$. By Ptolemy's Theorem, find the exact length of the other diagonal BD.

a. 10

b. $\frac{14}{5}$

c. $\frac{48}{5}$

d. 12

8 Cyclic quadrilateral $ABCD$ (shown schematically) has consecutive side lengths $AB = 8,\ BC = 15,\ CD = 8,\ DA = 15$, and diagonal $AC = 17$. By Ptolemy's Theorem, find the exact length of the other diagonal BD.

a. $\frac{161}{17}$

b. $\frac{240}{17}$

c. 19

d. 17

9 In the Cartesian plane, circle ω_1 has equation $x^2 + y^2 = 33$ and circle ω_2 has equation $(x - 4)^2 + y^2 = 33$. Their radical axis (the locus of points having equal power with respect to ω_1 and ω_2) is the vertical line $x = x_0$. Find x_0.

a. 1

b. 4

c. 2

d. 3

10 In the Cartesian plane, circle ω_1 has equation $x^2 + y^2 = 43$ and circle ω_2 has equation $(x - 6)^2 + y^2 = 31$. Their radical axis (the locus of points having equal power with respect to ω_1 and ω_2) is the vertical line $x = x_0$. Find x_0.

a. 3

b. 4

c. 1

d. 5

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11 In the Cartesian plane, circle ω_1 has equation $x^2 + y^2 = 57$ and circle ω_2 has equation $(x - 8)^2 + y^2 = 25$. Their radical axis (the locus of points having equal power with respect to ω_1 and ω_2) is the vertical line $x = x_0$. Find x_0.

a. 6

b. 4

c. 1

d. 7

12 In the Cartesian plane, circle ω_1 has equation $x^2 + y^2 = 75$ and circle ω_2 has equation $(x - 10)^2 + y^2 = 75$. Their radical axis (the locus of points having equal power with respect to ω_1 and ω_2) is the vertical line $x = x_0$. Find x_0.

a. 4

b. 10

c. 6

d. 5

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